



College	RV COLLEGE OF ENGINEERING®			
Department	COMPUTER SCIENCE AND ENGINEERING			
PROGRAM:	PG - CSE	Applied Artificial Intelligence and Machine Learning	Course Code	MCE202I
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Project Title	Optimum Racing: Real time race strategy prediction in formula one			
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Undertaken at	RV College of Engineering			

SYNOPSIS

Introduction:

Formula 1 (F1) is the pinnacle of motorsport, featuring high-performance, single-seater cars racing on circuits worldwide. Teams and drivers compete for the Constructors' and Drivers' Championships annually, with advanced engineering, strategy, and technology playing key roles.

The pitstop strategy includes deciding when and how many pit stops should be made in the given racing conditions.

Objectives:

To predict the race results in the form of driver rank and optimal pitstop strategy, which can be helpful for team principals and engineers to make critical pitstop decisions for winning.

The model would be able to optimize when a team should pit during a race based on real-time data, improving overall race strategy and reducing pit stop time.

Methodology:

We aim to implement time-series forecasting on real-time data from the OpenF1 API. The model will be trained on race data from 2011 to mid-2024, with the remainder of the 2024 season used for testing.

Key features include:

- Lap-by-Lap Data: Telemetry, tire compounds, pit stop timings.
- Derived Metrics: Tire degradation rates, position changes post-pit, traffic and gap patterns.
- Track Factors: Surface characteristics and their impact on performance.

We will employ transformer-based models (e.g., Chronos, TimeSFM) for multivariate predictions, comparing performance against traditional statistical models (LSTM, AutoRegressive Methods) to highlight advancements in forecasting accuracy and efficiency.



Data set used:

<https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020/data> For historical data

<https://www.f1cfa.com/> For Tyre Statistics

<https://openf1.org/> For real time Race Data

Software Requirements:

Programming Languages: Python

Data Processing: Numpy, Pandas

Data Visualization: Matplotlib, Seaborn

ML Frameworks: Scikit-learn, PyTorch, Chronos-FM

Version Control: Git

Hardware Requirements:

Processor: Intel i7/AMD Ryzen 5 or better (multi-core).

Memory: 8 GB RAM (minimum)

Storage: 512 GB SSD.

GPU: NVIDIA RTX 2060 or higher for deep learning models.

Contribution to the field:

Offer insights into competitor strategies by predicting their likely actions (e.g., pit stop windows or tire choices) and aid in simulating race scenarios for better pre-race preparation.

Open-source tools or dashboards that bring advanced analytics to commentators, fans, or amateur racing teams thus democratize insights traditionally locked within high-budget teams.

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